

Contact

301-758-5525 (Mobile)
mnpostol@gmail.com

www.linkedin.com/in/michael-postol (LinkedIn)

Top Skills

Topological Data Analysis
Anomaly Detection
Time Series Analysis

Publications

Algebraic Topology for Data Scientists

Michael Postol, Ph. D.

Principal Data Scientist and leading expert in Topological Data Analysis, with deep expertise in AI, Machine Learning, and Algorithm Design

Washington DC-Baltimore Area

Summary

Principal Data Scientist | Machine Learning | Topological Data Analysis | AI/ML for Cybersecurity | Anomaly Detection | Ph.D. Mathematics

Experienced data science leader with deep expertise in AI/ML, topological data analysis (TDA), and algorithm development. I bring over a decade of experience delivering innovative solutions in cybersecurity, time series classification, anomaly detection, and federated learning across government, defense, and research environments.

I specialize in applying advanced techniques—including algebraic topology, graph theory, and sheaf theory—to complex data challenges involving noisy, high-dimensional, and incomplete datasets. Skilled in Python (NumPy, Pandas, Scikit-learn, TensorFlow), cloud platforms (AWS, GCP), and scalable data processing workflows.

Author of "Algebraic Topology for Data Scientists," (<https://arxiv.org/pdf/2308.10825>) a leading online resource detailing the application of topological data analysis (TDA) to AI/ML, cybersecurity, and biomedical applications. Proven track record of leading cross-functional teams, driving technical strategy, and mentoring junior talent.

Looking to connect at mnpostol@gmail.com with organizations focused on AI innovation, data science leadership, and mission-driven technology.

Experience

MITRE

Principal Data Scientist

January 2022 - July 2025 (3 years 7 months)

McLean, Virginia, United States

Conducted research in AI/ML and data science for multiple public sector clients. Highlights include:

Authored Algebraic Topology for Data Scientists, a widely circulated online book introducing topological data analysis (TDA) for AI, cybersecurity, and biomedical applications.

Led applied research on sheaf theory for sensor fusion and data assimilation in the Bering Sea—marking the first known application of this mathematical technique to ocean science.

Developed AI readiness standards for U.S. Government datasets, influencing data governance strategies across agencies.

Contributed to a Concept of Operations document for a government financial system overhaul.

Designed advanced anomaly detection techniques for critical infrastructure and cyber event log data.

Applied federated learning and unsupervised techniques to event log and open-source malware-related PCAP data, enabling secure cross-departmental data sharing without compromising privacy.

U.S. Department of Defense Enterprise Discovery Operations Group

Senior Mathematician

September 2016 - December 2021 (5 years 4 months)

Fort George G. Meade, Maryland, United States

Highlights include:

Serving as Chair of Interagency Data Analytics Working Group.

Leading Academic partnership developing tools for time series classification and anomaly detection.

Developing novel techniques for using topological data analysis to classify multidimensional time series related to internet of things devices in a cyber-defense application.

U.S. Department of Defense National Threat Operations Center
Senior Mathematician

October 2014 - August 2016 (1 year 11 months)

Fort George G. Meade, Maryland, United States

Performed statistical and graphical analysis of infrastructure and timing behavior of malware implants.

Education

University of Minnesota

Doctor of Philosophy - PhD, Mathematics

University of Chicago

Bachelor of Science - BS, Mathematics